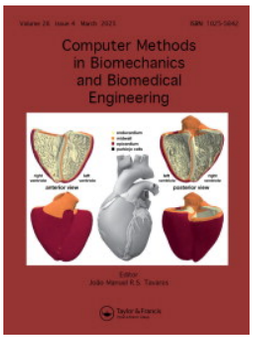

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Cognizance detection during mental arithmetic task using statistical approach

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ABSTRACT

The handheld diagnosis and analysis are highly dependent on the physiological data in the clinical sector. Detection of the defect in the neuronal-assisted activity raises the challenge to the prevailing treatment that benefits from machine learning approaches. The congregated EEG data is then utilized in design of learning applications to develop a model that classifies intricate EEG patterns into active and inactive segments. During arithmetic problem-solving EEG signal acquired from frontal lobe contributes for intelligence detection. The low intricate statistical parameters help in understanding the objective. The mean of the segmented samples and standard deviation are the features extracted for model building. The feature selection is handled using correlation and Fisher score between {Fp1 and F8} and priority ranking of the regions with enhanced activity are selected for the classifier models to the training net. The R-studio platform is used to classify the data based on active and inactive liability. The radial basis function kernel for support vector machine (SVM) is deployed to substantiate the proposed methodology. The vulnerable regions F1 and F8 for arithmetic activity can be visualized from the correlation fit performed between regions. Using SVM classifier sensitivity of 92.5% is obtained for the selected features. A wide range of clinical problems can be diagnosed using this model and used for brain-computer interface.

ARTICLE HISTORY
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KEYWORDS
EEG; correlation; kernel; SVM; patterns; R-studio

1 Introduction

The mammalian brain is made up of multiple neurons that are responsible for controlling the activity of the human body in response to intrinsic/extrinsic motor/sensory excitation. These nerve cells will transmit data from the human body to the brain. Analyzing signals or images from the brain can be used to better understand cognitive behavior. Eye motion, lip movement, recollection, attentiveness, hand grasping, and other motor and sensory states can be used to picture human behavior. These states are linked to a certain signal frequency, which aids in the comprehension of the functional activity of a complicated brain structure. EEG is a useful tool for obtaining brain waves from the scalp surface area that are correlated to different states. These signals are classed as δ, θ, α, β, and γ depending on signal frequencies in the range from 0.1 Hz to more than 100 Hz. The brain is an important part of the human body since it governs all of the body's functions. The brain can be thought of as a network of interconnected neurons that controls human behavior. Medical researchers are interested in learning more about the functional and cognitive activity of the human brain to develop better solutions to various brain-related ailments (Kumar and Bhuvaneswari 2012). According to medical physicians and radiologists, the left side of the brain controls the right side of the human body and vice versa. Electroencephalogram (EEG) uses the frequency of a signal to evaluate the brain and its actions. The brain is perhaps the most key functional organ in human body, as it commands and regulates all the muscles and nerves. The left half of the brain is known as left hemisphere and the other half on the right is known as a right hemisphere. There are four lobes in each hemisphere: frontal, temporal, parietal, and occipital. Each lobe controls and regulates various functions.

Arithmetical is a discipline of mathematics that deals with question-solving and numerical operations. Mathematical problem-solving involves a variety of cognitive processes and methods. To solve arithmetic problems, some mental operations are required: Data retention, information arrangement, and control on

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decision making. The diverse functions performed during arithmetic problem-solving are controlled by separate brain areas. The intraparietal sulcus participates consistently during numerical calculations (Peters and De Smedt 2018). Working memory is a general term for a brain function that is responsible for temporarily processing and storing information. It is critical for a variety of cognitive tasks. Several functional magnetic resonance imaging (MRI) investigations have discovered that executive function and working memory tasks engage the frontal lobe areas (Sarter et al. 1996). Working memory demand is thought to be supported by frontal lobe areas during mathematical activities. Varying types of task presentation result in different frontal hemisphere dominance. Visually presented activities tend to activate both the left and right frontal regions, but auditorily presented tasks activate the right frontal regions more.

In the recent studies conducted functional MRI (fMRI) data was used to study brain-behaviour correlation (Cheng et al. 2022). A study was conducted using brain scanning technique to analyze the arithmetic and metacognition (MC) of children. The study revealed the fact that prefrontal regions actively participated in the MC through empirical studies. EEG over other imaging modalities furnishes direct neural activity. The dependence of fMRI on blood oxygenation levels leads to indirect cognizant analysis (Gogna et al. 2021). Researchers from a range of domains, spanning from biophysical and connectomics to systems development and digital signal processing, have indicated an interest in investigating human cognitive function. An individual's brain activity patterns are influenced by emotional responses (Baars 2005). Cognitive processing emphasizes on the prediction of brain functioning during excited mental ability, sequential dynamic characteristics of the cortex, and interaction between neural centers.

EEG signals capture the electrical activity of the brain using electrodes attached to the scalp. The acquired signals are noisy, have artifacts. Doctors, neuroscientists, and biomedical engineers typically undergo years of training to comprehend and extract useful information from EEG data. Even in these circumstances, the raw collected data must be processed before being examined by specialists. Even if the person is stationary during recording, temporal and spatial filtering, as well as artifact rejection methods, are frequently used (Gemein et al. 2020). Clinical EEG analysis could be automated using machine learning (ML) techniques (Sharma et al. 2022). The data was separated into two categories: feature-based (custom elements) and end-to-end (with learned attributes). Visual inspection is a time-consuming, costly, and inconvenient technique. Therefore, does not scale well, and it cannot be used in brain-computer interface (BCI) applications. To automate, extend, and improve EEG data processing, low intricate AI and ML techniques are the ideal companions. To accomplish this a simple mathematical approach is proposed in this article to study the cognizance of human beings. Indeed, BCI (Roy 2022a, 2022b) systems like spellers and brain-controlled devices are built on decoding pipelines that employ a variety of machine-learning methods which can utilize the proposed model. The article primarily focuses on comparing traditional classification methods with deep learning-based techniques, specifically the Multi-Layered Perceptron (MLP) model. Support Vector Machine (SVM) showed the shortest training time and prediction speed with comparable accuracy among traditional ML-based algorithms. The subject-independent generalized MLP model successfully classified signals with approximately 90% accuracy and half the classification time compared to traditional ML-based models. This suggests that the MLP model can serve as a framework for advanced subject-independent BCI with optimization. The paper also mentions continuous efforts to improve deep learning-based BCI, such as the use of Deep Neural Autoencoders with sparsity constraints to extract hidden features from EEG data for subject-independent noninvasive BCI (Sharma et al. 2022). The article proposes an efficient multi-scale convolutional neural network (MS-CNN) for motor imagery (MI) classification in brain-machine interfaces using EEG signals. The MS-CNN model extracts distinguishable features from non-overlapping canonical frequency bands of EEG signals at multiple scales. Discriminant user-specific features are extracted and integrated into the CNN classifier to improve accuracy and performance. Data augmentation methods are implemented to further enhance the accuracy and robustness of the model. The proposed MS-CNN model achieves an average classification accuracy of 93.74% and Cohen's kappa-coefficient of 0.92 on the BCI competition IV2b dataset, outperforming several baseline and state-of-the-art EEG-based MI classification models. The MS-CNN model can be utilized in medical and healthcare applications, particularly for the detection and diagnosis of mental diseases and neurodegenerative diseases such as Parkinson's disease (Roy 2022b). The article proposes an efficient transfer learning-based multi-scale feature

fused CNN (MSFFCNN) for MI classification using electroencephalogram (EEG) signals. The MSFFCNN captures distinguishable features of various non-overlapping canonical frequency bands of EEG signals from different convolutional scales for multi-class MI classification. The article presents four different model variants, including subject-independent and subject-adaptive classification models, to account for inter-subject variability. Each adaptation configuration has been fine-tuned in an extensively trained pretrained model, and the performance of the classifier has been studied for a vast range of learning rates and degrees of adaptation. The proposed MSFFCNN achieves an average classification accuracy of 94.06% and outperforms several baseline and current state-of-the-art EEG-based MI classification models with fewer training samples. The article provides an effective and efficient transfer learning-based end-to-end MI classification framework for designing a high-performance robust MI-BCI system (Roy 2022a).

2. Materials and METHODS

2.1. Data acquisition, retrieval, and pre-processing

PhysioNet is a growing resource of well-characterized digital formats of physiological data and associated data for biomedical researchers to use. A web-based resource intended to supplement current research and drive future explorations into intricate physiologic and clinical data. The MIT Laboratory for Computational Physiology manages openly available medical research data. It is user-friendly. Using this repository, data from the database "EEG during mental arithmetic tasks" has been extracted in "edf" format.

The Neurocom EEG 23-channel equipment was used to capture the EEGs monopolarly. As shown in Figure 1, the surface electrodes were applied to the scalp with International 10/20 electrode reference. The interconnected ear reference electrodes were used to reference all electrodes.

- Impedance—>5 kΩ
- Sample rate—500 Hz
- HPF—cutoff frequency—0.5 Hz
- LPF—cutoff frequency—45 Hz
- Notch filter for powerline—50 Hz
- Time invariant of the amplification path—0.3 s
- Inter- conductive impedance—>5 kΩ (Dhanagopal et al. 2022).

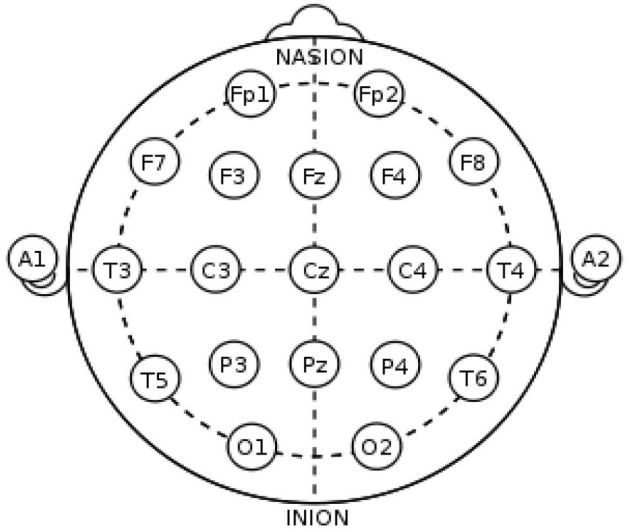


Figure 1. Electrodes of International 10-20 scheme for EEG (Khandpur 2014).

ML algorithm: Independent-Component-Analysis (ICA) is performed for removal of artifacts during the data pre-processing stage. EEG is a random signal and only specific segments contribute for the attainment of objective. The ICA dimensionality reduction helps to find relevant linear and nonlinear dependencies both EEG and noise parameters in the signal. The matrix multiplication of signals enables non-Gaussian distribution which aids in generating independent EEG signals that are noise free. The arithmetic objective was to subtract two numbers serially (Looi et al. 2016). All tests began with the verbalized communication of four-digit number subtracted from two-digit numbers. Each recording contains 180 s of artifact-free resting state EEG and cognition numeration EEG segments for 60 s. A total of 36 healthy volunteers' EEG recordings were acquired in two separate sets in the resting and active states.

A total of 66 individuals were initially signed up. Thirty patients were not included in the study because of the poor EEG quality.

Table 1 represents the subject information with age, gender, count quality, and the number of subtractions. This table gives an insight regarding efficient understanding of task done by the participants selected. The split of data acquired to various classes can be handled using this demarcation. The concluding sample size was 36. Participants were considered eligible if regular or rectified visual acuity, normal color perception were found, and no medical signs and symptoms of higher-level cognitive dysfunction, and no linguistic or gestural learning disabilities were reported. Psychotropic medication use, alcohol or

Table 1. Sample subject information.

Subject sample_ID	Age	Gender	Recording year	Number of subtractions	Quality count
Sample 00	21	F	2011	9.7	0
Sample 01	18	F	2011	29.35	1
Sample 02	19	F	2012	12.88	1
Sample 03	17	F	2010	31	1
Sample 04	17	F	2010	8.6	0
Sample 05	16	F	2010	20.71	1
Sample 06	18	M	2011	4.35	0
Sample 07	18	F	2012	13.38	1
Sample 08	26	M	2011	18.24	1
Sample 09	16	F	2010	7	0
Sample 10	17	F	2010	1	0
Sample 11	18	F	2010	26	1
Sample 12	17	F	2010	26.36	1
Sample 13	24	M	2012	34	1
Sample 14	17	F	2010	9	0
Sample 15	17	F	2012	22.18	1
Sample 16	17	F	2010	11.59	1
Sample 17	17	F	2010	28.7	1
Sample 18	17	F	2010	20	1
Sample 19	22	M	2010	7.06	0
Sample 20	17	F	2010	15.41	1
Sample 21	20	F	2010	1	0
Sample 22	19	F	2011	4.47	0
Sample 23	16	F	2010	27.47	1
Sample 24	17	M	2012	14.76	1
Sample 25	17	M	2012	30.53	1
Sample 26	17	F	2010	13.59	1
Sample 27	19	F	2010	34.59	1
Sample 28	19	F	2010	27	1
Sample 29	19	M	2011	16.59	1
Sample 30	17	M	2011	10	0
Sample 31	19	F	2012	19.88	1
Sample 32	20	F	2011	13	1
Sample 33	17	M	2010	21.47	1
Sample 34	18	F	2010	31	1
Sample 35	17	F	2010	12.18	1

drug dependency, and psychological or neurodegenerative diseases were all ruled out.

2.2. Data analysis

The data downloaded from PhysioNet is analyzed using an EDF analyzer in MATLAB. Various features of a signal like physical dimension, min max value, prefilter, number of samples are analyzed. Specific signal channel is visualized among 21 channels. The EEG data has to be differentiated based on relevant amplitude variations (Varshney et al. 2021). The temporal lobe performance significantly contributes for the above-mentioned parametric changes. From the above it is noted that Fz channels contributes more to arithmetic task and it has higher classification accuracy. Based on this analysis channels from frontal regions of brain Fp1, Fp2, F3, F4, F7, F8 were selected. These signals were specifically analyzed for each subject based on the active and inactive state. Figure 2 shows the EEG signal of subject_1 under rest state and active state of the same subject is given in Figure 3 while performing mental arithmetic task. Those signals are recorded at a single stretch and the acquired signals can be separated by two different bounded boxes as shown in Figure 4.

2.3. Statistical feature extraction from EEG

The extracted raw EEG signals for a subject at rest state and active state are shown in Figures 5–10 are converted into a numerical feature. EEG is a stochastic signal, so to deal with the random signal (Ma et al. 2021) statistical features like mean, variance, and standard deviation (SD) have been selected. A set of 72 observations were obtained for mean, variance, and SD which comprises active and inactive regions (Gurumurthy et al. 2013).

To select the optimum feature set Correlation coefficient and Fisher's score are used. Correlation coefficient is a statistical factor used to determine how close two variants are having a linear relation. Features with a high correlation are more linearly dependent and vice versa.

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}} \quad (1)$$

where r denotes Pearson Correlation Coefficient, n

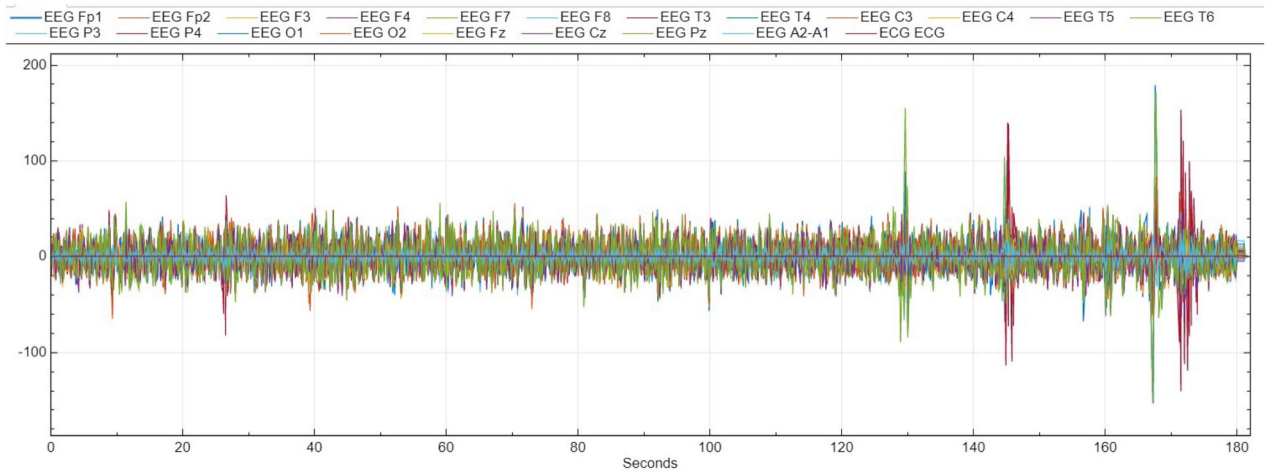


Figure 2. EEG signal of subject_1 under rest state.

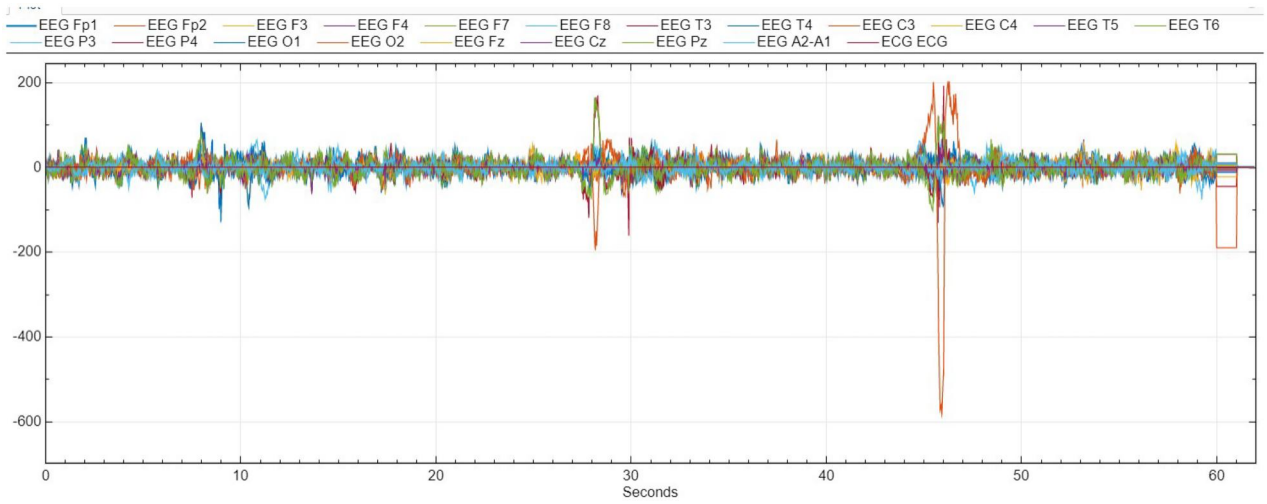


Figure 3. EEG signal of subject_1 while performing mental arithmetic task.

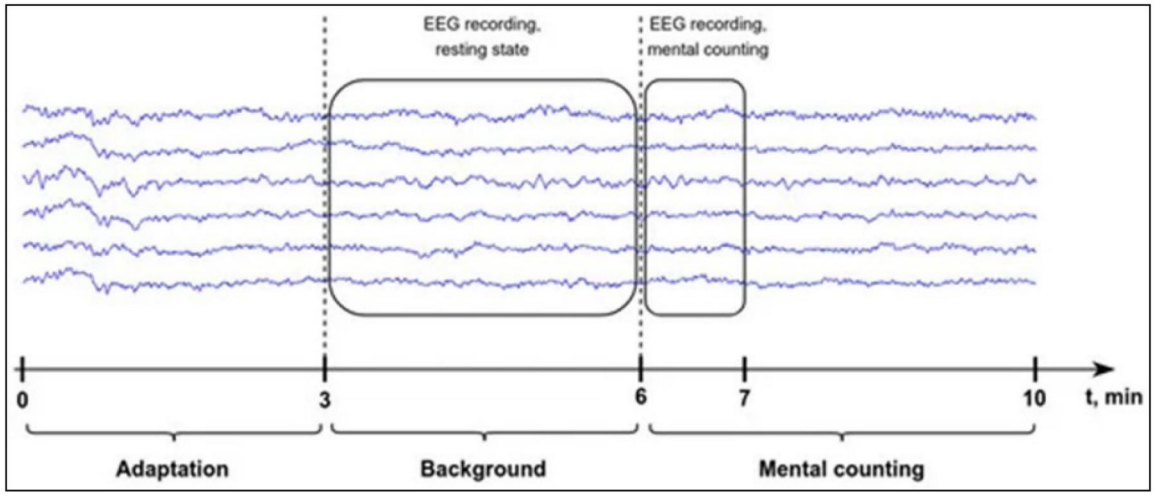


Figure 4. The two bounding boxes indicate the data that is available in the data set in two separate recording files.

represents the number of the sample pairs, $\sum xy$ denotes paired scores with sum of products, $\sum x$ gives the sum of the x scores, $\sum y$ gives the sum of the y scores in correlation, $\sum x^2$ denotes the sum of the squared x scores, and $\sum y^2$ denotes the sum of the squared y scores. A sub-optimal subset of features is

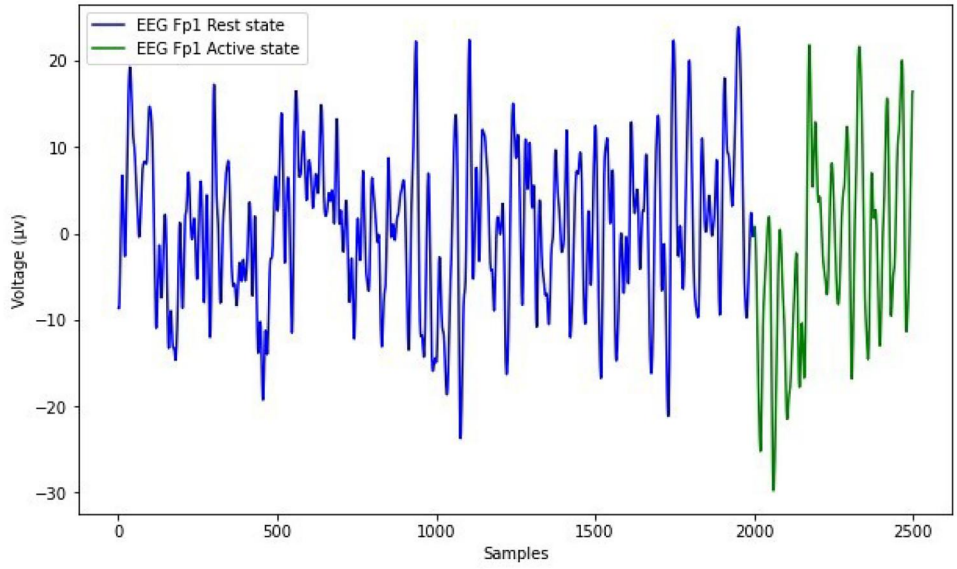


Figure 5. EEG Fp1—Rest and active state.

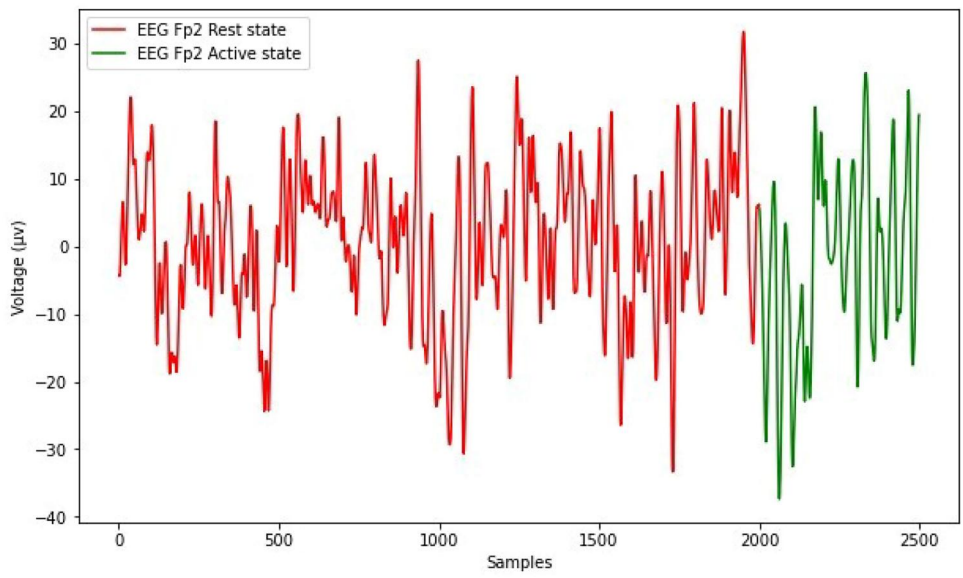


Figure 6. EEG Fp2—Rest and active state.

produced by Fisher's score, a rank-based feature selection approach that chooses each feature based on its Fisher criterion scores independently.

$$S_i = \frac{\sum n_j(\mu_{ij} - \mu_i)^2}{\sum n_j * \rho_{ij}^2} \tag{2}$$

ρ_{ij} denotes the variance of the ith feature in the jth class, μ_{ij} represent the mean of the ith feature in the jth class, μ_i denotes the mean of the ith feature, and n_i represent the number of instances in the jth class. Using RStudio various channels have been correlated to find correlation factor and p value (RStudio Team 2020). Based on the Correlation coefficient and Fisher's score best features have been extracted from Fp1, Fp2, F3, F4, F7, and F8.

2.4. Model description

Classification is a predictive modelling activity wherein a class label is predicted for a specific example of input data. It is a supervised learning technique that uses training data to determine or characterize the category of new set of observations. Classification is when a software learns from a dataset or observations and then categorizes subsequent

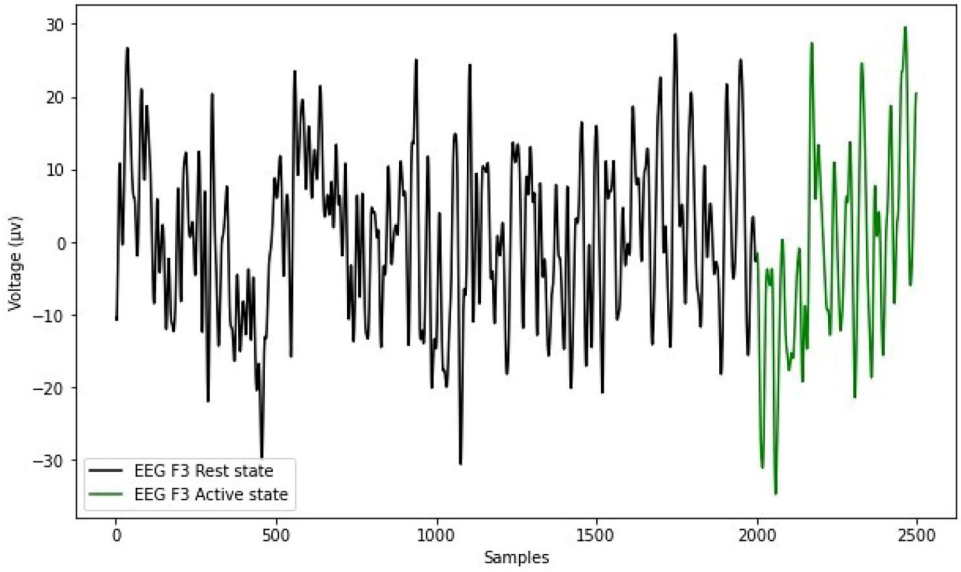


Figure 7. EEG F3—Rest and active state.

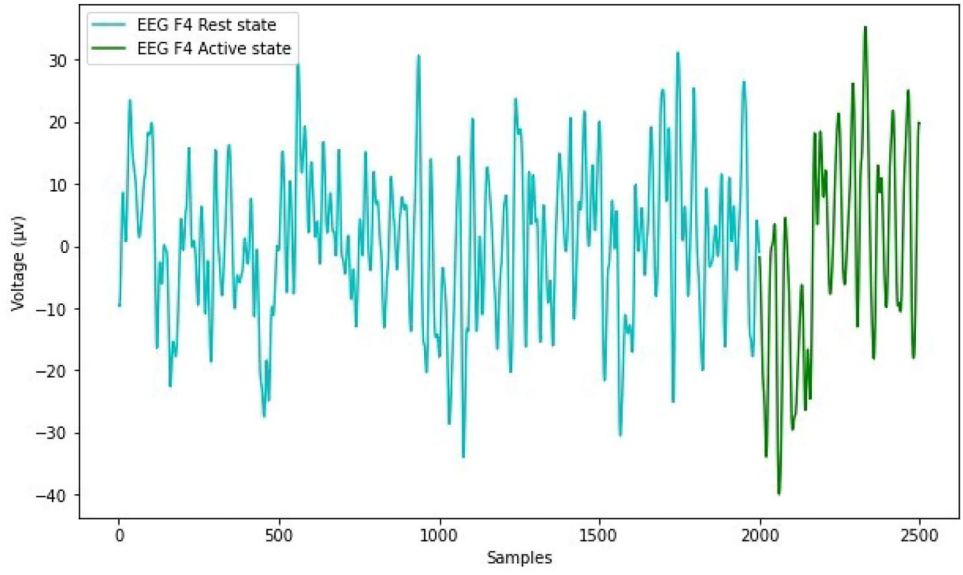


Figure 8. EEG F4—Rest and active state.

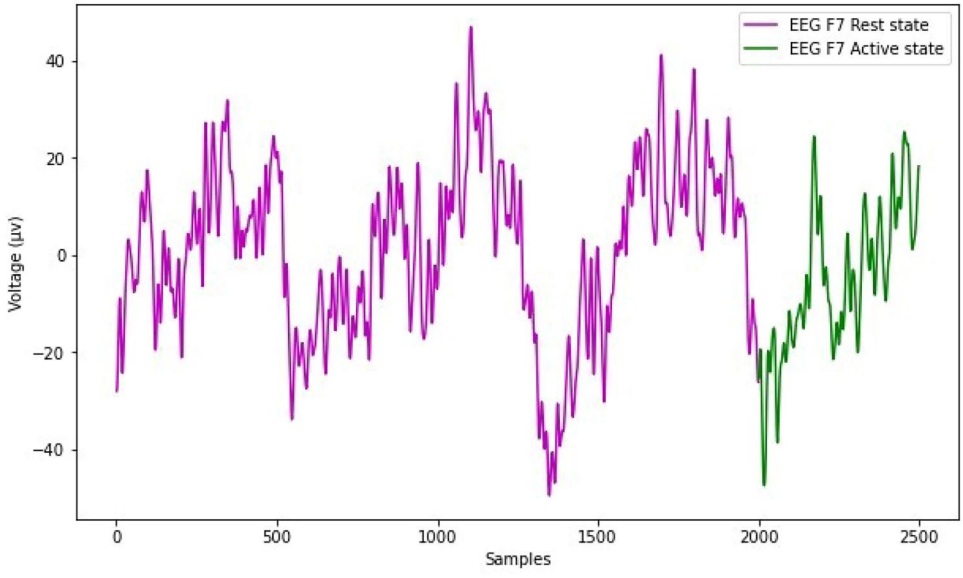


Figure 9. EEG F7—Rest and active state.

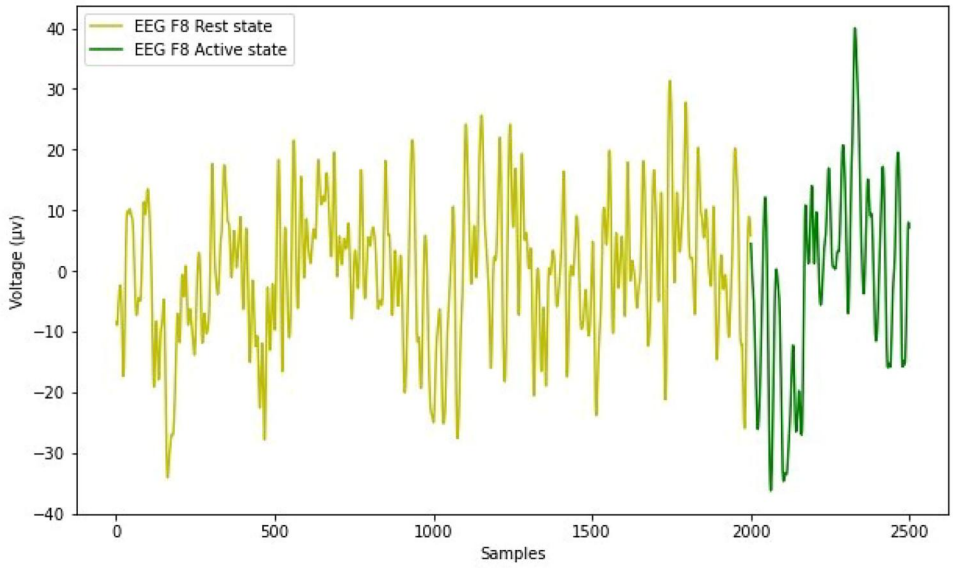


Figure 10. EEG F8—Rest and active state.

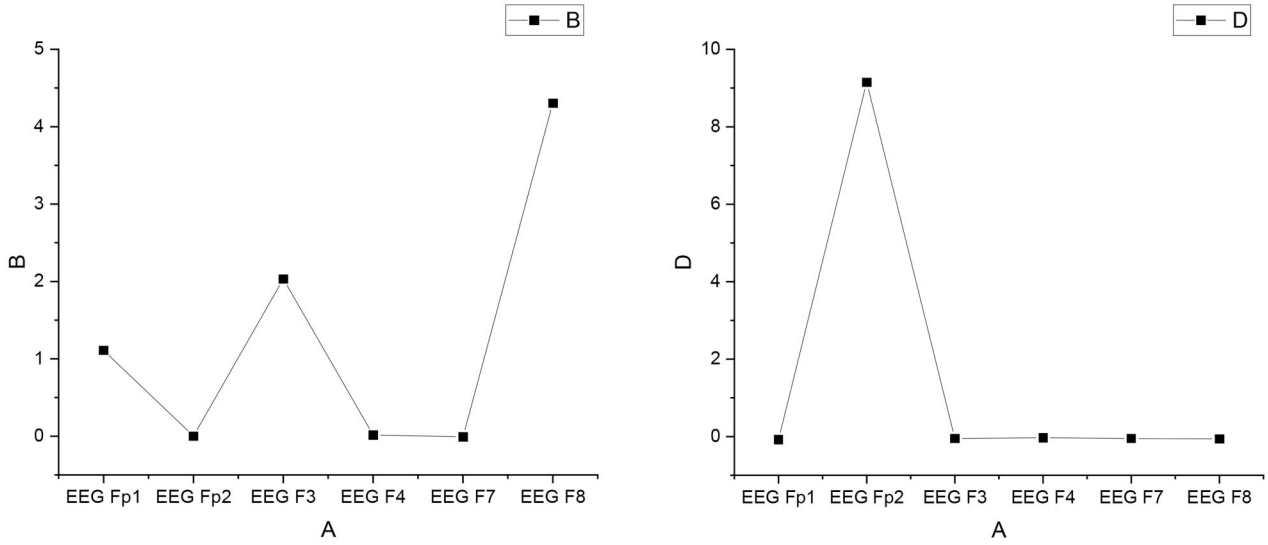


Figure 11. Line plot of nominal mean obtained from six channel systems during inactive state and active state respectively.

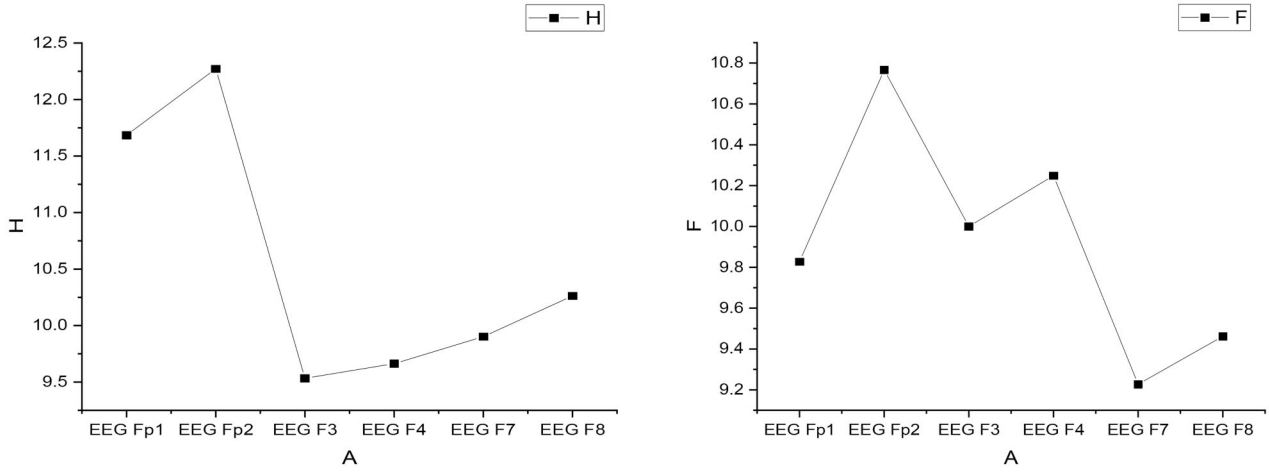


Figure 12. Line plot of nominal standard deviation obtained from six channel systems during inactive state and active state respectively.

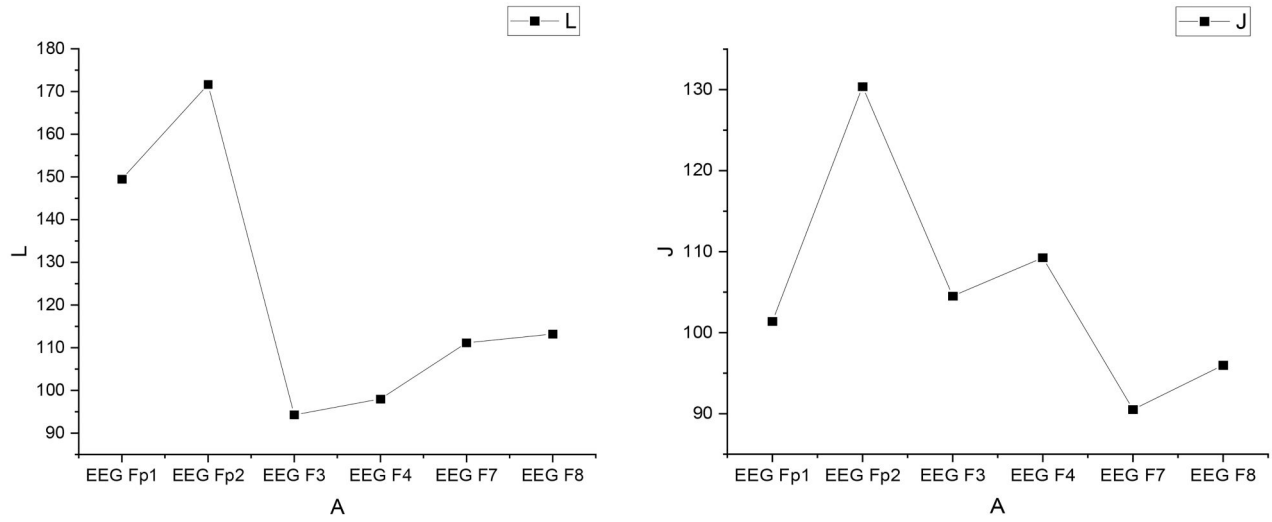


Figure 13. Line plot of average variance obtained from six channel systems during inactive state and active state respectively.

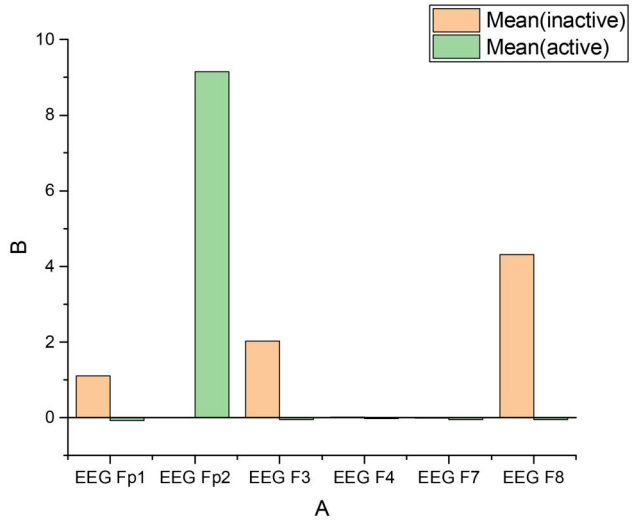


Figure 14. Bar chart of comparison between nominal mean during active and inactive state.

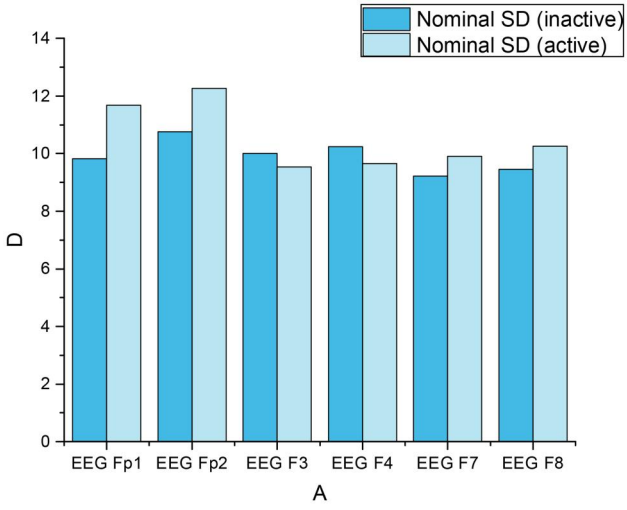


Figure 15. Bar chart of comparison between nominal standard deviation during active and inactive state.

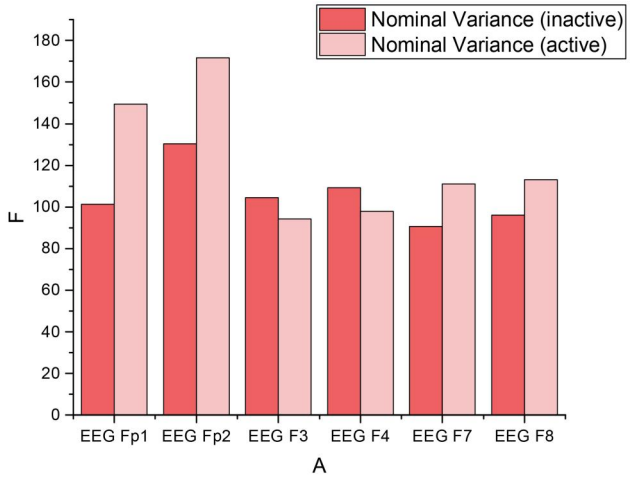
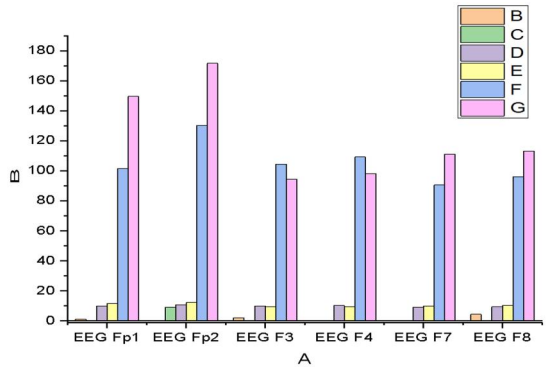


Figure 16. Bar chart of comparison between nominal variance during active and inactive state.

observations into one of a number of classes or groups (Murtazina and Avdeenko 2020). It enables us to break down massive volumes of data into discrete values like zero or one, proven or disproven, or a predefined outcome label class.

SVM algorithm has proven to be an effective method for resolving practical binary classification problems (Fatimah et al. 2020). SVM has been proved to outperform other supervised learning approaches. SVM is a robust classifier which is capable of performing classification using generated support vectors (SV). The SV creates a linearly separable hyperplane using kernel trick methodology. The decision boundary is generated using the various mathematical formulations pertaining to the linear, polynomial and Gaussian models. SVM have been one of the most widely used classification methods in recent years, owing to its solid theoretical basis and high generalization capacity. The SVM method is used to find



B - Mean of inactive signal
C - Mean of active signal

D - Standard deviation (SD) of inactive signal
E - SD of active signal

F - Variance of inactive signal
Variance of active signal

X-axis - Channels
Y-axis - Frequency of the channels

Figure 17. Comparison of mean, SD, and variance of frontal region channels.

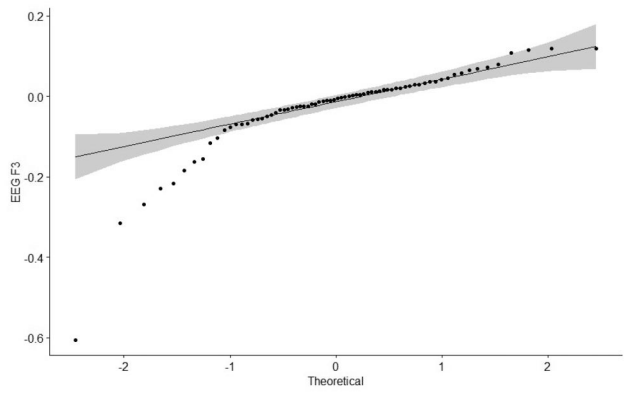


Figure 18. Correlation graph of EEG F3.

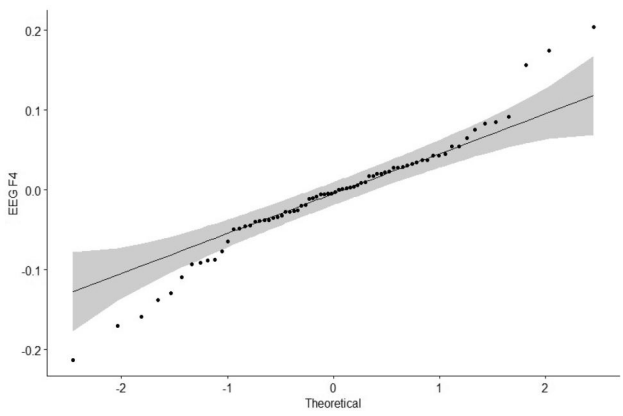


Figure 19. Correlation graph of EEG F4.

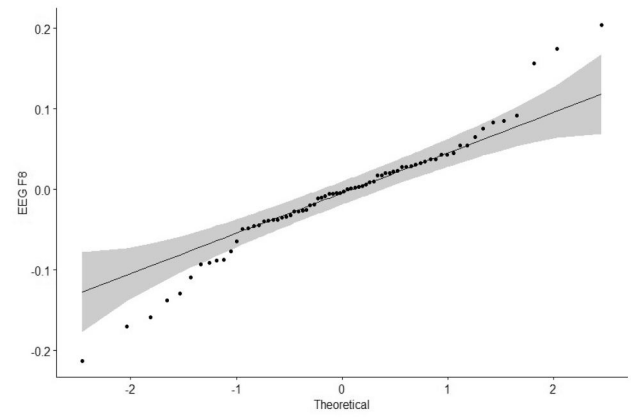


Figure 20. Correlation graph of EEG F8.

the optimum variable or decision boundary for categorizing an n-dimensional region into classes, enabling the following data sets to be simply placed in the appropriate category (Wang and Sourina 2013). The idealistic decision-making boundary is best known as a hyperplane. SVM is used to choose the extreme arguments that help build the hyperplane. SV are the names given to these data points manifested using statistical features.

Using RStudio, SVM classification was carried out for different types of kernels. The extracted features were imported and it was classified into active and inactive segments by following steps.

- Import the dataset in ".csv" format
- Install necessary libraries and Feature Scaling
- Encode the target feature as a factor
- Split the dataset into a training set and a test set.
- Fit SVM to Training set
- Predict the Test set results
- Extract Confusion Matrix
- Plot the training data set results

These sequential steps were followed to classify the EEG signals into active and inactive segments.

3. Results and discussion

The statistical analysis with the parameters namely means SD and variance provide us with insights on the performance of different channels. The analysis of nominal values of all the parameters gives the activity range from each channel. The difference in its frequency is observed with the statistical features. The

Table 2. Correlation factors of six channels.

Correlation values	EEG Fp1	EEG Fp2	EEG F3	EEG F4	EEG F7	EEG F8
EEG Fp1	1	−0.02174251	0.9997063	0.08540895	0.04919494	−0.01269195
EEG Fp2	−0.02174251	1	−0.01734126	−0.04329104	−0.2867294	−0.01664872
EEG F3	0.9997063	−0.01734126	1	0.07424206	0.2020726	−0.01302035
EEG F4	0.08540895	−0.04329104	0.07424206	1	0.1403227	0.1403227
EEG F7	0.04919494	−0.2867294	0.0491455	0.2020726	1	0.1403227
EEG F8	−0.01269195	−0.0166487	−0.01302035	0.1403227	0.004943199	1

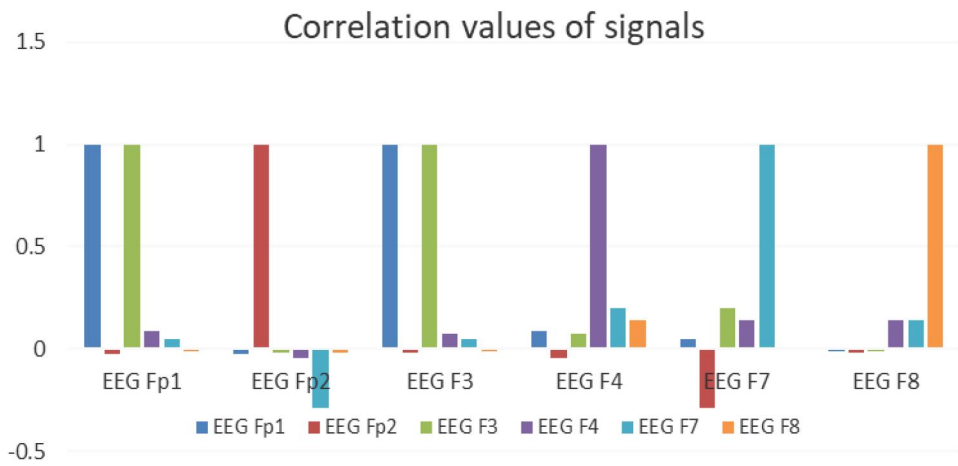


Figure 21. Correlation graph of six channels.

Table 3. Fisher's score of six channels.

Fisher's score	EEG Fp1	EEG Fp2	EEG F3	EEG F4	EEG F7	EEG F8
EEG Fp1	2.20E-16	0.8561	2.20E-16	0.4756	0.6815	0.9157
EEG Fp2	0.8561	2.20E-16	0.885	0.718	0.01461	0.8896
EEG F3	2.20E-16	0.885	2.20E-16	0.5354	0.6818	0.9136
EEG F4	0.4756	0.718	0.5354	2.20E-16	0.08871	0.2397
EEG F7	0.6815	0.01461	0.6818	0.08871	2.20E-16	0.9671
EEG F8	0.9157	0.8896	0.9136	0.2397	0.9671	2.20E-16

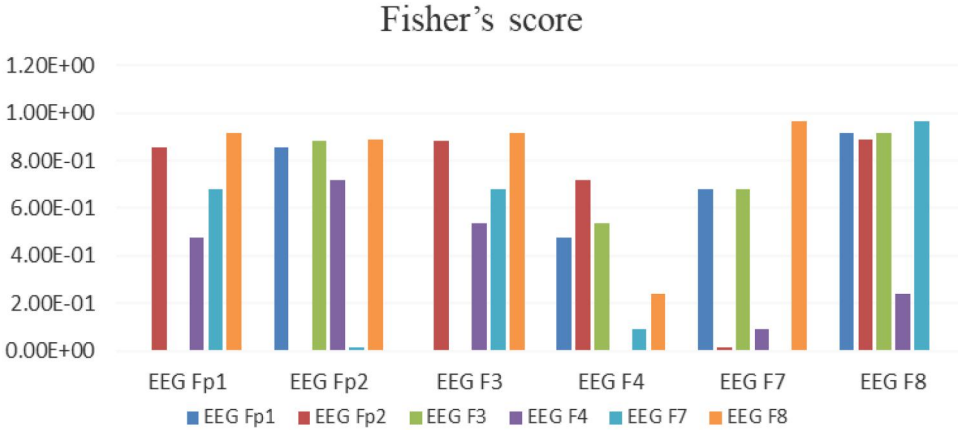


Figure 22. Fisher's score graph of six channels.

mean values observed from the bar graph suggests having more activity on the Fp1 and Fp2 channel, which is the frontal lobe region (Zyma et al. 2019). The SD and variance follows the same trend. This analysis of extracted features cannot be used for the classification methods since this is a crude analysis. No prior normalization has been done with the acquired frequency values, hence analysis with such values is not reliable for any conceptual inferences. Linear plot of nominal mean, SD, and variance are

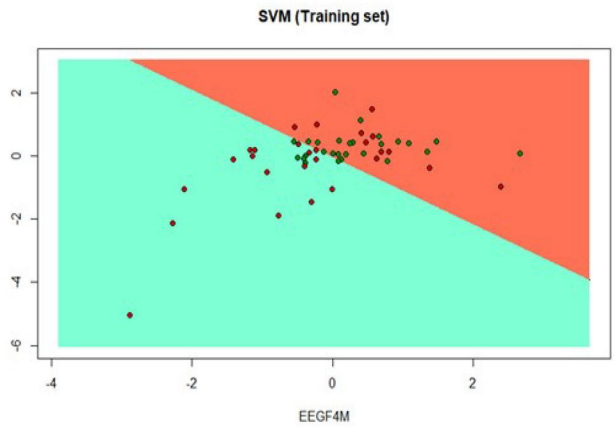


Figure 23. SVM classification for EEG F8 and EEG F4 with linear kernel.

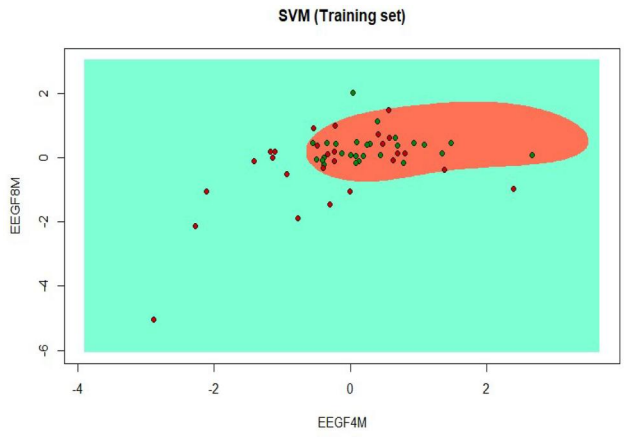


Figure 24. SVM classification for EEG F8 and EEG F4 with radial kernel.

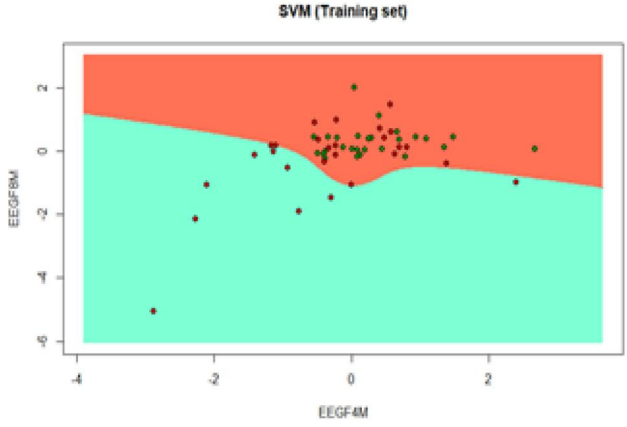


Figure 25. SVM classification for EEG F8 and EEG F4 with polynomial kernel.

plotted shown in Figures 11–13. Bar chart compares the active and inactive regions of the frontal lobe which is shown in Figures 14–17. EEG Fp1 and EEG Fp2 shows significant difference which can be analyzed from these graphs.

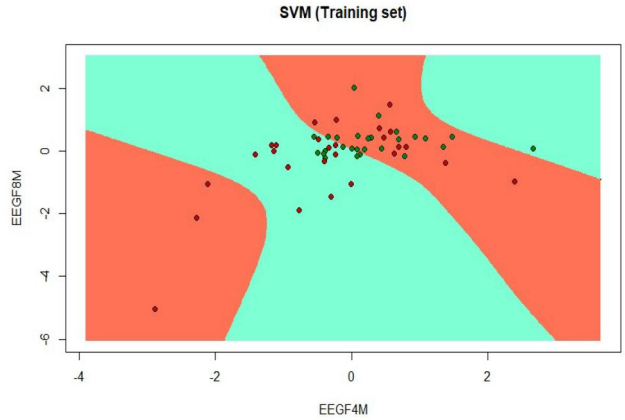


Figure 26. SVM classification for EEG F8 and EEG F4 with sigmoid kernel.

The feature selection requires the dimension of the data labels to be reduced. This involves various techniques. The technique employed is the filter method. Filter method used here are correlation and Fisher's score. These methods are designated for feature selection process. The correlation factor from the graph gives positive correlation for the channels F3, F4, and F8 shown in Figures 18–20. Table 2 shown below tabulates all the correlation factors against each channel. The correlation graph shown in Figure 21 depicts that the channel F3, F4, and F8 have positive correlation values compared to others. The Fisher's score values were also calculated which are tabulated in Table 3. F3, F4, and F8 have higher scores which can be analyzed using the plot in Figure 22. The Fisher's score values match the result of the correlation coefficients. Hence F3, F4, and F8 are chosen as prospective features. These channels are from the frontal lobes respectively. The dimension is thus reduced from 6×72 to 3×72. The features selected can be employed in classification for more accurate results.

The classification is done using SVM binary algorithm. This requires two variables which are separated as class 0 and class 1. The features are F4 and F8, which have been further reduced in accordance with the facts from the literature. The F4 and F8 features are used for classification with various kernels namely linear, radial, sigmoid, and polynomial which are shown in Figures 23–26. The use of kernels can accommodate more SV into certain classes. The linear kernel provided distinctive classification across the hyperplane. The projections in different dimensions give other kernels. The radial kernel has larger accuracy which implies more demarcation of feature vectors.

SVM is a data classification algorithm that offers an accuracy (Gogna et al. 2023) of 86.1%, due to its

data that can be linearly and nonlinearly separated. SVM also reduces the risk of over-fitting. The exactness of the accuracy is given by precision. Recall Sensitivity of 92.5% tell us that it is good at avoiding false alarms. The harmonized average of recall and precision is the F1 score. It gives a validation on the accuracy percentage. Radial basis function is the most popular type of kernel function. It is localized and it has a finite output along the entire x-axis since it is localized. The numerical product between two points in an extremely appropriate feature space is returned by the kernel functions.

- Accuracy = 86.1%
- Precision = 75.7%
- Recall sensitivity = 92.5%
- Specificity = 82.2%
- F1 score = 83.2%

SVM classifiers perform better than the Naive Bayes approach in terms of prediction accuracy and speed. A work carried out for concentration (Bird et al. 2018) aided EEG classification produced an accuracy of 56.3% for correlation-based dataset using Naïve Bayes classifier. Since a sample of training instances is used, the decision stage requires less memory. SVM performs well because it has a vast dimensional space and a distinct separation margin. ML trade-offs can be seen in SVM and kNN. kNN classifier for mental workload using 5-fold classes (Cheema et al. 2018) produces an accuracy of 61.43% of accuracy. This model also incorporates optimization of features selected using statistical parameters. SVM can only recognize a small subset of patterns, but it is simpler to grasp and requires significantly less compute than kNN. On the other hand, even while kNN may find incredibly complicated patterns, its output is more challenging to understand.

4. Conclusions

The EEG data helps us fetching insights regarding the thinking ability of an individual. This motto is achieved using Electroencephalogram visual imagery data in the work proposed in this article. The learning algorithm used to check this capability utilizes a minimum feature for immediate interpretation. The use cases of the algorithm are for a wider horizon. The number of channels for executing mental arithmetic task categorization for best accuracy was implemented using a linear classifier. The moderation in the kernels used also intended to improvise the performance. The methodology emphasizes that the performance of mental arithmetic exercises greatly relies on the frontal lobe excitatory activity rather than the other data collection points. The model described despite of the lower computational complexity is not utilized for real time datasets which helps understanding the nonlinear complexities exhibited by the EEG signal. In the future works module for IoMT(Internet of Medical Things)-based EEG acquisition and validation can be implemented. People who have lost their ability to process cognitive activity will benefit significantly from this algorithm which is used to validate tasks with minimal features. The BCI allows the users to communicate and control external devices (Beniczky et al. 2021) can also be designed using this simple more efficient model. Using an assorted signal would allow the user to interact with the system better. IQ levels, concentration analysis (activity in the prefrontal cortex) can be diagnosed using trained models of EEG (Yadava et al. 2017). The ML classification models can be exploited in studying the activity of patients with autism and plays a major role in neuro-marketing (Goldberger et al. 2000).

Disclosure statement

The authors declare that there are no conflicts of interest regarding the publication of this article.

Data availability

The data are available in the following link: https://archive.physionet.org/cgi-bin/atm/ATM.

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